Omnisequence: A Preliminary Framework for Omnisequential Arithmetic and Cosmology

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Abstract — This paper introduces the Omnisequence, a mathematical framework for describing infinite succession, predecessor-successor relationships and continuously evolving states, motivated by the questions surrounding cosmological expansion and the tendency of many mathematical and physical systems to progress through states. This paper defines the Omnisequence as a foundational object and developed concepts of Omnisequential athematic and Omnisequential cosmology, Definitions, Axioms, Operations. examples along with open questions and areas for future development. The goal of this paper is not to provide a completed theory, but to form a foundation from witch the framework may b refined and investigated further

Keywords — Omnisequence, Succession, Predecessor, Cosmology, Expansion.

Notation:
O: Omnisequence
O_{loc}: Local View
O_{glo}: Global View
$\uparrow$: Increment
$\downarrow$: Decrement
NS_n: Output State Label

I. INTRODUCTION

THE notion of succession appears in numerous contexts,from natural numbers, to cosmological evolution.

Natural numbers as defined by [1], all natural numbers have a successor: $S(n)$. The successor of 1 is 2, the successor of 3 is 4 and so on.

Succession also appears in biological systems, life succeeds and evolves to a next state with each state arising from what came before it.

A similar patters can be observed in cosmology. The universe as we understand it grows and succeeds its previous state even if it is by a small amount causing time to always move forward as a result, For a seemingly infinite time period.

This raises the question: "If Sucsessorship is infinite that must mean Predecessorship is also infinite", A potential objection arises, cosmological models describe a extremely hot and dense early state of the universe, a point infinitely dense that contains all space and in conjuncture all time. meaning anything beyond that dense point did not exist as for it to exist it would have to be in said point.

But Since the universe is expanding it is reason to believe that it has always been expanding, never actually having a final predecessor or the most simplest state. Existing notions of infinity are extremely useful, but they do not directly describe the type of self-succession explored in this paper.

II. THE OMNISEQUENCE

Figure 1: As shown in the image this is the symbol representation of the Omnisequence. For typing simplicity we will use the capital letter "O"

To investigate these questions, the Omnisequence is a proposed as an ordered structure intended to model infinite succession and infinite Predecessorship simultaneously, unlike traditional sequences the Omnisequence is not defined by a first element followed by repeated applications of a successor function. instead, every state in O has a unique successor $\&$ predecessor.

II.1 Core Axioms

Successor — For any state "x" in O there exists a unique successor $S(x)$.

$$\forall x| \in O,\ \exists S(x) \tag{1}$$

Predecessor — For any state "x" in O there exists a unique predecessor $P(x)$.

$$\forall x| \in O,\ \exists P(x) \tag{2}$$

Every state must have a successor, likewise ever state must have a predecessor. Successor and Predecessor are inverse operations.

$$S(P(x)) = x \quad \& \quad P(S(x)) = x \tag{3}$$

II.2 Local and Global Views

The Omnisequence may be examined from two distinct perspectives, while the Omnisequence as a whole is continuously succeeding and changing, the values in it remain observable through recurring predecessor-successor transitions. to distinguish between the two perceptive the concepts of local and global views are introduced

II.2.1 Local view

Local view, denoted by O_{loc} considers the values contained within the omniscience, although individual occurrences of a value may transition into successor states, the predecessor-successor chain ensures that the same value continues to exist within the Omnisequence. for example, while a occurrences of 1 may transition into 2 another occurrences of 0 simultaneously transitions into 1, in result every value that exists remain present within O_{loc}.

II.2.2 Global View

The Global view, denoted by O_{glo} considers the Omnisequence as a complete structure, with this view every value is continuously succeeding into another value and the Omnisequence itself is never static. rather than focusing on the values present. O_{glo} focuses on the overall evolution and self-succeeding behavior

II.2.3 Local - Global Axioms

State Equality — The Omnisequence is defined as a continuously succeeding structure consequently, equality must be considered with respect to the state being observed:

$$O_{glo}(t) = O_{glo}(t) \tag{4}$$

$$O_{glo}(t) \neq O_{glo}(t + \Delta t) \tag{5}$$

III. OMNISEQUENTIAL ARITHMETIC

While the local and global view describe how the Omnisequence might be observed they don't describe how transitions occur within the states, Omnisequential arithmetic introduces the concepts of increments

III.0.1 Increments

A increment records a transition from one state to another within the Omnisequence returning every possible value it takes to get from one state to another. positive increments are denoted by the symbol $\uparrow$ and decrement are notated by $\downarrow$

unlike classical arrhythmic, a increment does not only represent a change in value instead it records the transition itself including the successor-predecessor relationship that produced the resulting state
Eg.

$$O + 1 = \uparrow 2 \tag{6}$$

This indicates the base state was 1 and that the Omnisequence returned the successor state 2. The symbol $\uparrow$ records the transition leading to that state.

III.1 Repeated Increments

Additional increments may be applied to a previously incremented result. Each new increment repairs the total effect of all previous increments

for a single increment,

$$\uparrow 2 \tag{7}$$

the total incremental value is 1

for 2 increments,

$$\uparrow\uparrow 2 \tag{8}$$

the second increment repeats the effects of the first, making the total incremental value of 2

for three increments

$$\uparrow\uparrow\uparrow 2 \tag{9}$$

the third incrementation represents the combine effects of the previous two increments, producing a total increment of 4 meaning the total value of this equation is 2 with all possibility's of incrementing to 4. Instead of writing each arrow its also accseptable to write the number of incrementation present underneeth the arrow as shown in the image below..

$$\underset{n}{\uparrow}\underset{n}{\downarrow}$$

Figure 2: As shown in the image this is the symbol representation of increment and decrements with a incremental value of n. For typing simplicity we can also refer to the incremental value as the subscript of the increment: $\uparrow_n$

The output of an Omnisequential operation can be denoted by a new state, abbreviated as NS
Given a state NS_n the subscript identify the successor generation of the state

Note this notation only serve as labels for successor generations and does not alter the value contained in the state

III.2 Increment Axioms

Existence — Every increment must originate from a state within the Omnisequence
Direction — Positive increments are represented by $\uparrow$ and decrement represented by $\downarrow$
Recursion — each additional increment repeats the cumulative effect of all preceding increments
Closure — Applying an increment to a state of the Omnisequence produces another valid state within the Omnisequence
Increment Inversion — Increments and decrements are inverse operations, applying a increment followed by a equivalent decrement returns the original state. and vice vera. formally:

$$\uparrow^n x + \downarrow^n x = O \;\;\&\;\; \downarrow^n x + \uparrow^n x = O \tag{10}$$

consequently, every successor transition possesses a corresponding predecessor transitions capable of reversing its effect

Eg.

$$O + 1 = \uparrow 2 \tag{11}$$

Applying another increment gives

$$\uparrow\uparrow 2 \tag{12}$$

The first increment contributes 1. The second increment repeats the total effect of the first, their fore the $\sum$ incremental value is 2 and the resulting possible state being 3 likewise,

$$\uparrow\uparrow\uparrow 2 \tag{13}$$

Has a $\sum$ incremental value of 4 and resulting in possible state of 5

Incrementation between similar states — A increment requires a transition between distinct states, if a base state and the state its reaching is the same, then there cannot increment, for example:

$$O + O = 2O \tag{14}$$

This would also align with the "State Equality" Axiom, while the equation below would not..

$$O + O = \uparrow O \tag{15}$$

Successor Priority — Omnisequential outputs are shown as successor transitions, Decrements ($\downarrow$) may be used as a inverse operation within Omnisequential arithmetic.

Incremental Acquisition — When a natural number without incrementation is combined with a incremental state, the natural number contributes to its magnitude as additional increments to the resulting state. for example:

$$\uparrow 17 + 4 \; = \uparrow_5 17 \tag{16}$$

$$\uparrow 3 + 2 \; = \uparrow_3 3 \tag{17}$$

Incremental Consolidation — When two incremental states are combined, the incremental value does not change. instead the terminal states combine while preserving a single incremental designation. For example:

$$\uparrow 5 + \uparrow 5 \; = \uparrow_5 10 \tag{18}$$

$$\uparrow 13 + \; \uparrow 4 \; = \uparrow 17 \tag{19}$$

IV. OMNISEQUENTIAL COSMOLOGY

The Omnisequence was originally motivated by the question concerning the origin and evolution of the universe. while Omnisequential arithmetic provides a framework for describing succession, Omnisequential cosmology exploratory possibility that succession might be a fundamental property of reality.

A analogy fro the Omnisequence may be found in fractal structures. When viewed from a distance, a fractal can appear to posses a definite boundary or endpoint. Increasing the level of magnification often reveals additional self similar structure. what first appears to be a end point was instead a limitation of observation

The Omnisequence provides a similar perspective, regarding predecessor chains. A state might appear fundamental or initial when viewed at a limited scale. However under the Omnisequential framework, evert state possesses a Predecessor and successor. beginnings may reflect observational limits rather that actual boundary of the structure

> *This analogy is only a illustration of Omnisequential viewpoint and should not be interpreted as evidence that physical reality is a fractal*

REFERENCES

[1] Betrans Russle F.R.S. Alfread North Whitehead, *Principia mathematica*, Cambrage University Press, 1910.